OSPF Questions

13 questions | Original English questions and source answers

Question 1

You have configured a router with an OSPF router ID, but its IP address still reflects the physical interface. Which action can you take to correct the problem in the least disruptive way?

Options

A. Save the router configuration

B. Reload the router

C. Reload the OSPF process

D. Specify a loopback address

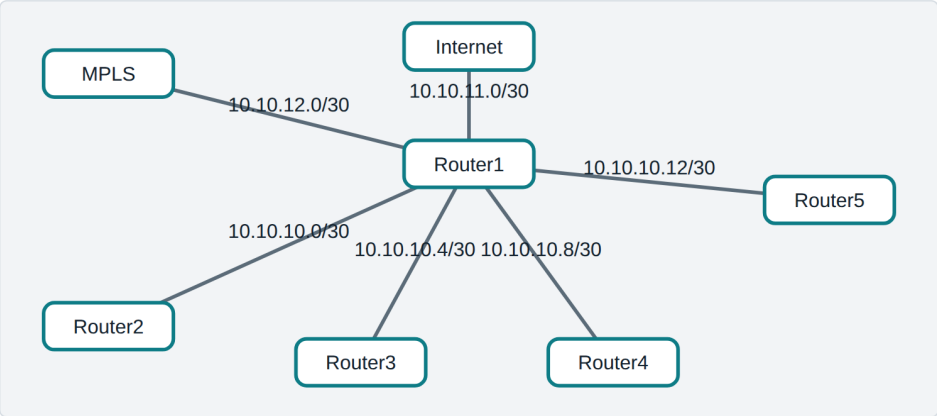
Answer

C. Reload the OSPF process

Source: CCNA 200-301 Questions.html | Question 1

Question 2

Refer to the exhibit. To which device does Router1 send packets that are destined to host 10.10.13.165?




```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
      209.165.200.0/27 is subnetted, 1 subnets
B        209.165.200.224 [20/0] via 10.10.12.2,03:32:14
      209.165.201.0/27 is subnetted, 1 subnets
B        209.165.201.0 [20/0] via 10.10.12.2,02:26:53
      209.165.202.0/27 is subnetted, 1 subnets
B        209.165.202.128 [20/0] via 10.10.12.2,02:46:03
      10.0.0.0/8 is variably subnetted, 10 subnets, 4 masks
O        10.10.13.0/25 [110/2] via 10.10.10.1,00:00:04, GigabitEthernet0/0
O        10.10.13.128/28 [110/2] via 10.10.10.5,00:00:12, GigabitEthernet0/1
O        10.10.13.144/28 [110/2] via 10.10.10.9,00:01:57, GigabitEthernet0/2
O        10.10.13.160/29 [110/2] via 10.10.10.5,00:00:12, GigabitEthernet0/1
O        10.10.13.208/29 [110/2] via 10.10.10.13,00:01:57, GigabitEthernet0/3
S*    0.0.0.0/0 [1/0] via 10.10.11.2
```

Options

A. Router2

B. Router5

C. Router3

D. Router4

Answer

C. Router3

Source explanation

The destination of 10.10.13.165 matches the entry “O 10.10.13.160/29 [110/2]...” because of the longest prefix length rule so the packet will be forwarded to 10.10.10.5, which is Router3.

Source: CCNA 200-301 Questions.html | Question 2

Question 3

Refer to the exhibit. What does router R1 use as its OSPF router-ID?

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES NVRAM  administratively down down
GigabitEthernet1/0     192.168.0.1     YES NVRAM  up                    up
GigabitEthernet2/0     10.10.1.10      YES NVRAM  up                    up
GigabitEthernet3/0     10.10.10.20     YES NVRAM  up                    up
GigabitEthernet4/0     unassigned      YES NVRAM  administratively down down
Loopback0              172.16.15.10    YES manual up                    up
```

Options

A. 10.10.10.20

B. 192.168.0.1

C. 10.10.1.10

D. 172.16.15.10

Answer

D. 172.16.15.10

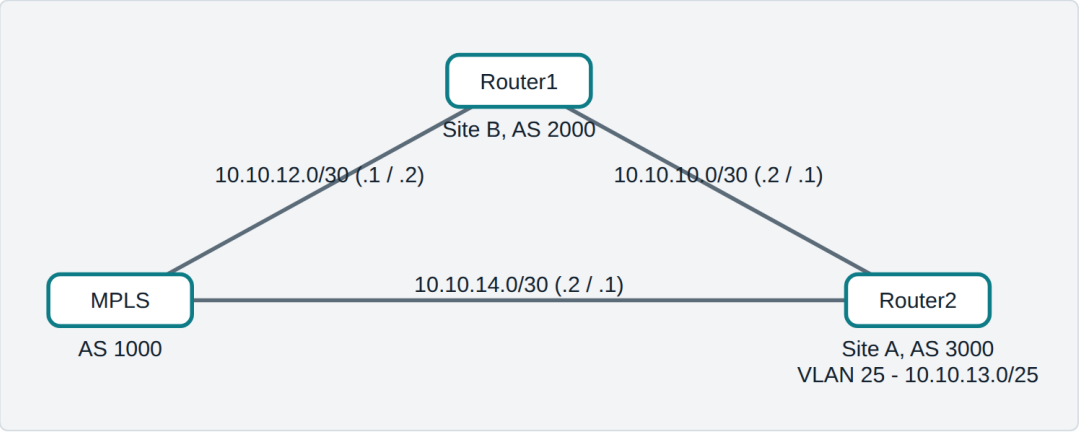
Source explanation

1. Manual configuration of the router ID (via the “router-id x.x.x.x” command under OSPF router configuration mode).
2. Highest IP address on a loopback interface.
3. Highest IP address on a non-loopback and active (no shutdown) interface.

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Refer to the exhibit. An engineer is bringing up a new circuit to the MPLS provider on the Gi0/1 interface of Router1. The new circuit uses eBGP and learns the route to VLAN25 from the BGP path.
What is the expected behavior for the traffic flow for route 10.10.13.0/25?




```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
      10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C        10.10.10.0/28 is directly connected, GigabitEthernet0/0
C        10.10.11.0/30 is directly connected, FastEthernet2/0
O        10.10.13.0/25 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.128/28 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.144/28 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.160/29 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.208/30 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
O        10.10.13.252/30 [110/2] via 10.10.10.1,00:00:32, GigabitEthernet0/0
S*    0.0.0.0/0 [1/0] via 10.10.11.2
```

Options

A. Route 10.10.13.0/25 learned via the Gi0/0 interface remains in the routing table

B. Traffic to 10.10.13.0.25 is load balanced out of multiple interfaces

C. Traffic to 10.10.13.0/25 is symmetrical

D. Route 10.10.13.0/25 is updated in the routing table as being learned from interface Gi0/1

Answer

D. Route 10.10.13.0/25 is updated in the routing table as being learned from interface Gi0/1

Source explanation

From the output we can deduce interface Gi0/0 on R1 is the interface connecting to Router2 so interface Gi0/1 is the interface connecting to the new MPLS AS 1000.

The AD of eBGP (20) is smaller than that of OSPF (110) so the route to 10.10.13.0/25 will be updated as being learned from the new BGP path.

Source: CCNA 200-301 Questions.html | Question 4

Question 5

A user configured OSPF in a single area between two routers. A serial interface connecting R1 and R2 is running encapsulation PPP. By default which OSPF network type is seen on this interface when the user types show ip ospf interface on R1 or R2?

Options

A. point-to-multipoint

B. non-broadcast

C. point-to-point

D. broadcast

Answer

C. point-to-point

Source explanation

The default OSPF network type for HDLC and PPP on Serial link is point-to-point (while the default OSPF network type for Ethernet link is Broadcast).

Source: CCNA 200-301 Questions.html | Question 5

Question 6

Refer to the exhibit. If OSPF is running on this network, how does Router 2 handle traffic from Site B to 10.10.13.128/25 at Site A?

```
Site A -- Router1 ======== Router2 -- Site B

Router2#show ip route
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.10.10.8/30 is directly connected, FastEthernet0/2
C        10.10.10.12/30 is directly connected, FastEthernet0/1
O        10.10.13.0/25 [110/11] via 10.10.10.9, 00:00:02, FastEthernet0/2
                       [110/11] via 10.10.10.13, 00:00:02, FastEthernet0/1
C        10.10.10.4/30 is directly connected, FastEthernet0/2
```

Options

A. It sends packets out of interface Fa0/1 only

B. It is unreachable and discards the traffic

C. It load-balances traffic out of Fa0/1 and Fa0/2

D. It sends packets out of interface Fa0/2 only

Answer

B. It is unreachable and discards the traffic

Source explanation

Router2 does not have an entry for the subnet 10.10.13.128/25. It only has an entry for 10.10.13.0/25, which ranges from 10.10.13.0 to 10.10.13.127.

Source: CCNA 200-301 Questions.html | Question 6

Question 7

Refer to the exhibit. The default-information originate command is configured under the R1 OSPF configuration. After testing, workstations on VLAN 20 at Site B cannot reach a DNS server on the Internet. Which action corrects the configuration issue?

```
Site B: R2 (.1) --10.10.10.0/30-- (.2) R1 :Site A (.17) --10.10.10.16/30-- (.18)
Internet

R2#show run | b router ospf
router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 auto-cost reference-bandwidth 10000
 network 10.10.10.1 0.0.0.0 area 0
 network 10.10.13.1 0.0.0.0 area 0

R1#show run | b router ospf
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 auto-cost reference-bandwidth 10000
 network 10.10.10.2 0.0.0.0 area 0
 default-information originate

R2#show ip route
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C        10.10.10.0/30 is directly connected
C        10.10.13.0/25 is directly connected, Vlan20
C        10.10.13.144/28 is directly connected, Vlan40

R1#show ip route
Gateway of last resort is not set
      10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C        10.10.10.0/30 is directly connected, FastEthernet0/1
O        10.10.13.0/25 [110/5766] via 10.10.10.1,00:33:22
C        10.10.10.16/30 is directly connected, FastEthernet0/24
O        10.10.13.144/28 [110/110] via 10.10.10.1, 01:33:33
```

Options

A. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.18 command on R1

B. Add the always keyword to the default-information originate command on R1

C. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.2 command on R2

D. Add the default-information originate command on R2

Answer

A. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.18 command on R1

Source explanation

The default-information originate command advertises a default route into a normal area, provided the advertising router already has a default route. But in the routing table of R1 we don’t see such a default route. Therefore we have to configure one.

We cannot only add the "always" keyword to the command "default-information originate" on R1 because R2 still does not know how to forward traffic received from R1.

Source: CCNA 200-301 Questions.html | Question 7

Question 8

An engineer configured an OSPF neighbor as a designated router. Which state verifies the designated router is in the proper mode?

Options

A. Exchange

B. Full

C. 2-way

D. Init

Answer

B. Full

Source: CCNA 200-301 Questions.html | Question 8

Question 9

What is the purpose of the show ip ospf interface command?

Options

A. displaying OSPF neighbor information on a per-interface-type basis

B. displaying general information about OSPF routing processes

C. displaying OSPF-related interface information

D. displaying OSPF neighbor information on a per-interface basis

Answer

C. displaying OSPF-related interface information

Source: CCNA 200-301 Questions.html | Question 9

Question 10

R1 has learned route 192.168.12.0/24 via IS-IS, OSPF, RIP and Internal EIGRP. Under normal operating conditions, which routing protocol is installed in the routing table?

Options

A. RIP

B. IS-IS

C. OSPF

D. Internal EIGRP

Answer

D. Internal EIGRP

Source explanation

With the same route (prefix), the router will choose the routing protocol with lowest Administrative Distance (AD) to install into the routing table. The AD of Internal EIGRP (90) is lowest so it would be chosen. The table below lists the ADs of popular routing protocols.

| Routing protocol | AD

| Directly Connected | 0

| Static | 1

| EIGRP | 90

| EIGRP Summary route | 5

| OSPF | 110

| RIP | 120

Note: The AD of IS-IS is 115. The “EIGRP” in the table above is “Internal EIGRP”. The AD of “External EIGRP” is 170. An EIGRP external route is a route that was redistributed into EIGRP.

With the same route (prefix), the router will choose the routing protocol with lowest Administrative Distance (AD) to install into the routing table. The AD of Internal EIGRP (90) is lowest so it would be chosen. The table below lists the ADs of popular routing protocols.

Routing protocol	AD
Directly Connected	0
Static	1
EIGRP	90
EIGRP Summary route	5
OSPF	110
RIP	120

Note: The AD of IS-IS is 115. The “EIGRP” in the table above is “Internal EIGRP”. The AD of “External EIGRP” is 170. An EIGRP external route is a route that was redistributed into EIGRP.

Source: CCNA 200-301 Questions.html | Question 10

Question 11

Refer to the exhibit. The “show ip ospf interface” command has been executed on R1. How is OSPF configured?

```
Designated Router (ID) 10.11.11.11, Interface address 10.10.10.1
Backup Designated router (ID) 10.3.3.3, Interface address 10.10.10.3
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:08
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
...
Last flood scan time is 0 msec, maximum is 1 msec
Neighbor Count is 3, Adjacent neighbor count is 3
  Adjacent with neighbor 10.1.1.4
  Adjacent with neighbor 10.2.2.2
  Adjacent with neighbor 10.3.3.3 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
```

Options

A. The interface is not participating in OSPF

B. A point-to-point network type is configured

C. There are six OSPF neighbors on this interface

D. The default Hello and Dead timers are in use

Answer

D. The default Hello and Dead timers are in use

Source explanation

From the output we can see there are Designated Router & Backup Designated Router for this OSPF domain so this is a broadcast network (point-to-point and point-to-multipoint networks do not elect DR & BDR) -> Answer 'A point-to-point network type is configured' is not correct.

By default, the timers on a broadcast network (Ethernet, point-to-point and point-to-multipoint) are 10 seconds hello and 40 seconds dead (therefore answer 'The default Hello and Dead timers are in use' is correct). The timers on a non-broadcast network are 30 seconds hello 120 seconds dead.

From the line “Neighbor Count is 3”, we learn there are four OSPF routers in this OSPF domain -> Answer 'There are six OSPF neighbors on this interface' is not correct.

Source: CCNA 200-301 Questions.html | Question 11

Question 12

When OSPF learns multiple paths to a network, how does it select a route?

Options

A. It multiple the active K value by 256 to calculate the route with the lowest metric.

B. For each existing interface, it adds the metric from the source router to the destination to calculate the route with the lowest bandwidth.

C. It divides a reference bandwidth of 100 Mbps by the actual bandwidth of the existing interface to calculate the router with the lowest cost.

D. It count the number of hops between the source router and the destination to determine the router with the lowest metric

Answer

C. It divides a reference bandwidth of 100 Mbps by the actual bandwidth of the existing interface to calculate the router with the lowest cost.

Source: CCNA 200-301 Questions.html | Question 12

Question 13

A user configured OSPF and advertised the Gigabit Ethernet interface in OSPF. By default, which type of OSPF network does this interface belong to?

Options

A. nonbroadcast

B. broadcast

C. point-to-point

D. point-to-multipoint

Answer

B. broadcast

Source explanation

The Broadcast network type is the default for an OSPF enabled ethernet interface (while Point-to-Point is the default OSPF network type for Serial interface with HDLC and PPP encapsulation).

Reference: https://www.oreilly.com/library/view/cisco-ios-cookbook/0596527225/ch08s15.html

Source: CCNA 200-301 Questions.html | Question 13